

ALAMOGORDO WHITE SANDS, NM, USA

WMO: 722693

Lat: 32.840N

Lon: 105.991W

Elev: 1280

StdP: 86.86

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 93097

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.2	-3.9	-18.8	0.8	5.8	-17.4	0.9	7.4	11.1	11.4	9.4	11.9	1.6	140	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	37.7	17.3	37.0	17.4	35.1	17.2	21.3	30.0	20.4	29.2	19.8	28.6	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	16.1	23.4	17.9	15.1	23.2	17.5	14.6	23.2	68.3	29.0	65.1	28.9	62.8	28.7	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.5	7.5	-10.2	39.8	4.0	1.2	-13.0	40.6	-15.3	41.4	-17.6	42.1	-20.4	43.0		
(5)			-11.2	21.8	3.4	1.0	-13.6	22.6	-15.6	23.2	-17.5	23.7	-19.9	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	6.3	8.7	12.6	16.6	21.2	26.5	27.1	26.3	23.3	16.9	10.4	6.1	(6)
(7)		DBStd	8.39	3.69	4.23	4.00	3.68	3.42	2.57	2.14	2.13	2.68	4.09	4.08	3.96	(7)
(8)		HDD10.0	394	124	66	22	2	0	0	0	0	0	4	45	132	(8)
(9)		HDD18.3	1603	372	269	178	74	14	0	0	0	2	74	239	379	(9)
(10)		CDD10.0	2901	10	29	103	199	346	493	529	505	400	218	56	11	(10)
(11)		CDD18.3	1068	0	0	2	21	102	243	271	247	152	30	1	0	(11)
(12)		CDH23.3	14127	0	7	107	460	1577	3481	3387	2928	1714	438	29	0	(12)
(13)		CDH26.7	7164	0	0	13	110	719	2021	1856	1542	783	120	1	0	(13)
(14)	Wind	WSAvg	3.0	2.6	3.1	3.4	3.8	3.6	3.4	3.0	2.7	2.6	2.8	2.5	2.6	(14)
(15)	Precipitation	PrecAvg	319	17	14	11	6	12	24	56	60	48	33	18	21	(15)
(16)		PrecMax	466	36	30	55	29	60	56	162	146	158	155	86	76	(16)
(17)		PrecMin	149	0	0	0	0	0	0	7	7	0	0	0	0	(17)
(18)		PrecStd	81	11	8	13	8	13	18	34	34	34	42	23	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	24.0	27.9	31.3	37.1	39.2	38.9	37.8	36.0	32.3	26.3	21.2	(19)
(20)			MCWB	9.0	9.6	11.3	12.9	16.2	17.1	17.5	18.2	18.0	16.5	12.7	10.1	(20)
(21)		2%	DB	18.7	22.3	26.2	29.0	34.0	37.8	37.5	36.4	34.0	29.1	23.8	18.7	(21)
(22)			MCWB	7.9	9.0	10.4	12.2	14.7	16.6	18.0	18.7	17.3	15.0	11.4	9.1	(22)
(23)		5%	DB	17.2	19.9	24.0	27.7	32.5	37.2	36.2	35.0	32.7	27.7	22.1	17.1	(23)
(24)			MCWB	7.3	8.2	9.7	11.6	14.2	16.4	18.3	18.3	17.0	14.3	10.8	7.6	(24)
(25)		10%	DB	14.8	17.7	22.5	26.3	31.0	35.1	34.1	33.7	31.4	26.1	19.0	14.0	(25)
(26)			MCWB	6.0	7.3	9.1	11.0	13.6	15.8	18.3	18.4	16.8	13.7	9.4	6.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	11.6	13.5	15.9	18.5	20.9	22.9	22.1	20.9	19.4	14.7	11.4	(27)
(28)			MCDWB	17.0	18.3	23.0	26.2	30.5	32.6	33.2	30.7	28.7	25.7	22.3	18.6	(28)
(29)		2%	WB	9.1	10.2	11.8	14.0	16.9	19.7	21.5	21.2	19.9	17.8	12.8	9.6	(29)
(30)			MCDWB	15.9	18.4	23.1	25.2	29.5	30.8	30.2	29.8	27.9	24.6	20.2	16.7	(30)
(31)		5%	WB	8.0	9.2	10.7	12.8	15.7	18.7	20.5	20.4	19.1	16.3	11.6	8.2	(31)
(32)			MCDWB	15.1	17.5	21.9	24.5	28.5	30.5	28.9	29.1	27.2	23.7	19.1	15.0	(32)
(33)		10%	WB	6.8	8.1	9.7	11.7	14.7	17.8	19.8	19.8	18.4	15.1	10.3	6.9	(33)
(34)			MCDWB	13.7	16.6	20.8	24.0	27.4	30.4	28.5	28.5	26.7	23.1	18.3	13.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.0	15.2	16.2	16.6	17.0	16.6	14.3	14.2	14.6	15.4	15.6	14.2	(35)
(36)			MCDBR	17.3	18.1	18.5	18.1	18.8	18.2	16.3	16.1	16.5	17.2	18.2	16.6	(36)
(37)			MCWBR	9.7	9.5	9.0	8.6	8.4	7.2	5.0	4.9	6.0	7.7	9.2	9.2	(37)
(38)		5% WB	MCDBR	15.0	15.5	17.0	16.9	17.3	15.9	13.6	13.6	13.3	13.7	15.3	14.6	(38)
(39)			MCWBR	8.9	8.6	8.6	8.5	7.8	6.3	4.5	4.7	5.0	6.5	8.4	8.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.251	0.259	0.279	0.307	0.324	0.348	0.389	0.372	0.334	0.291	0.270	0.256	(40)	
(41)		taud	2.606	2.579	2.512	2.417	2.424	2.390	2.349	2.368	2.438	2.552	2.561	2.593	(41)	
(42)		Ebn at Noon	992	1011	1006	981	959	931	891	905	931	958	957	966	(42)	
(43)		Edh at Noon	76	87	101	116	117	120	125	121	107	88	79	73	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.55	4.54	5.86	7.18	7.82	7.78	6.93	6.43	5.79	4.81	3.79	3.17	(44)	
(45)		RadStd	0.25	0.36	0.32	0.33	0.32	0.41	0.37	0.27	0.29	0.39	0.26	0.18	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page